BUSINESS ANALYSIS CASE STUDY · HUY HA · SOLE BA, SELF-DIRECTED

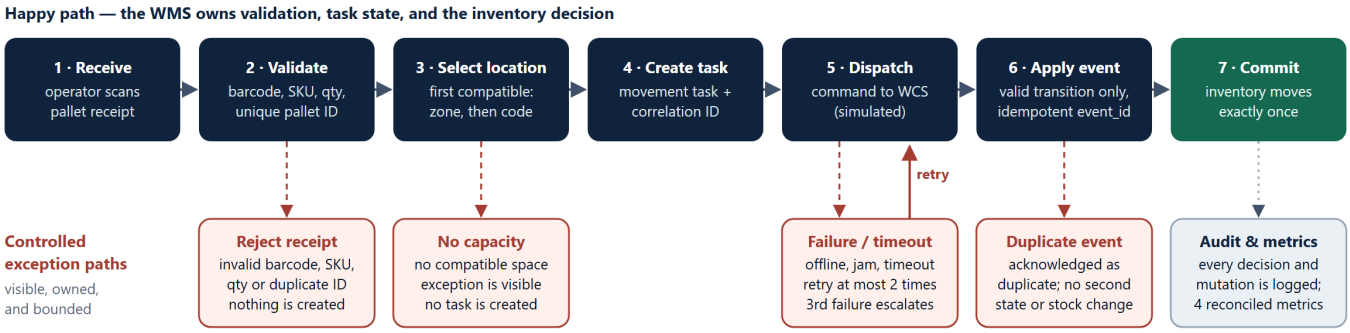
Smart Warehouse Inbound & Exception Management

Business Analysis case study · WMS/WCS inbound putaway

Live demo qahuy.github.io/smart-warehouse-inbound-exceptions/demo/
Full case study qahuy.github.io/smart-warehouse-inbound-exceptions/
Repository github.com/QaHuy/smart-warehouse-inbound-exceptions

> **Scope disclosure.** Personal WMS/WCS portfolio case study using synthetic data. WCS behavior and device events are simulated. No interview, site observation, measured benefit, or production system is claimed.

4	10 · 10 · 5	12 / 26	14 / 14	0
business needs	FR · BR · NFR	user stories / ACs	UAT cases passed	orphaned requirements



Simplified view of diagrams/to-be-inbound.bpmn. Synthetic data; WCS commands and device events are simulated. Cancellation from eligible states is omitted for readability.

PROBLEM

Inbound checks are fragmented and exception visibility is poor. Barcode, SKU, quantity, duplicate-pallet, capacity, and device checks happen at different points, so nobody can say with confidence whether a pallet is safe to put away, who owns a failed movement, or whether inventory was updated before the pallet actually arrived.

GOAL

A traceable pallet receipt and putaway flow with safe recovery: invalid pallets are stopped early, every WMS-to-WCS movement is traceable, exceptions are visible and bounded, and inventory changes **exactly once**, only after verified completion.

WHAT I DID

- Framed the problem as **4 business needs** and a fixed inbound-only scope; mapped **5 stakeholder personas** with a proposed RACI and a 45-minute interview guide.
- Modelled the assumed **As-Is** and proposed **To-Be** in **BPMN 2.0**, plus a system context, WMS–WCS sequence, and task **state machine**.
- Wrote an **SRS/FSD-lite**: 10 functional requirements, 10 business rules, 5 NFRs, and a 13-code error catalogue.
- Ordered a **12-story backlog** by dependency and operational risk; each story passes an INVEST check and has positive and negative **Given/When/Then** criteria (26 in total).
- Specified the **data and interface contracts**: 9-table SQLite schema with data dictionary, **OpenAPI 3.1** (8 operations), and 10 SQL control and metric queries.
- Validated the rules with a **clickable prototype** and **14 automated UAT cases**, linked end to end in a requirements traceability matrix (BN → FR → BR → US → UAT).

KEY DECISIONS

Deterministic putaway	First compatible location below capacity, ordered by zone priority then location code.
Completion-gated inventory	Inventory moves only on a valid, unique `TASK_COMPLETED` event.
Bounded retry	Maximum two retries; the third failure escalates to a person.
UTC ISO 8601 timestamps	All simulator timestamps use UTC ISO 8601.

RESULTS — AND WHAT IS NOT CLAIMED

- **14 / 14** automated UAT cases pass, including negative paths, duplicate events, and metric reconciliation.
- Traceability BN → FR → BR → US → UAT with **0 orphaned requirements**.
- Not claimed: interviews, site observation, measured benefit, or real device integration — next step is discovery with real users.

SKILLS DEMONSTRATED

Requirements elicitation · Stakeholder analysis & RACI · BPMN 2.0 · UML state & sequence diagrams · SRS / FSD · Business rules · User stories (INVEST) · Acceptance criteria (Given/When/Then) · Gap analysis · Requirements traceability matrix · UAT planning & execution · OpenAPI · SQL · Data dictionary · Agile backlog prioritisation · Change control